

For each of these circuits.

1. Calculate the total resistance in ohms.

$$R_T = R_1 + R_2 + R_3$$

$$20 + 30 + 50 = 100\Omega$$

2. Total Current from the Power Supply.

$$I_T = \frac{125}{100}$$

$$I_T = 1.25A$$

3. Current through each resistor.

$$I_{20\Omega} = I_{30\Omega} = I_{50\Omega} = 1.25A$$

In a series circuit, current is the same everywhere.

4. Voltage Drop Across each resistor.

$$V = IR$$

Across 20Ω $V_1 = 1.25 \times 20 = 25V$

Across 30Ω $V_2 = 1.25 \times 30 = 37.5V$

Across 50Ω $V_3 = 1.25 \times 50 = 62.5V$

$$25V + 37.5V + 62.5V = 125V$$

5. Power Dissipated by Each Resistor.

Using Watt's law

$$P = VI$$

$$P_1 = 25 \times 1.25 = 31.25W$$

Power in 20Ω

Power in 30Ω $P_2 = 37.5 \times 1.25 = 46.875W$

Power in 50Ω $P_3 = 62.5 \times 1.25 = 78.125W$

Parallel Circuit.

1. Total Resistance.

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\frac{1}{R_T} = 0.05 + 0.01 + 0.02$$

$$= 0.08 \Rightarrow 12.5 \Omega$$

2. Total Current from the Power Supply.

$$I_T = \frac{V}{R_T}$$

$$I_T = \frac{125}{12.5} = 10A$$

3. Current through each resistor

Using $I = \frac{V}{R}$

current through 20Ω $I_1 = \frac{125}{20} = 6.25A$

$$100\Omega = I_2 = \frac{125}{100}$$

$$I_2 = 1.25A$$

$$50\Omega = I_3 = \frac{125}{50}$$

$$I_3 = 2.5A$$

$$6.25A + 1.25 + 2.5 = 10A$$

4. $V_1 = V_2 = V_3 = 125V$

5. Power dissipated through each resistor

$$P = VI$$

$$\text{Power in } 20\Omega \quad P_1 = 125 \times 6.25 = 781.25W$$

$$\text{Power in } 100\Omega \quad P_2 = 125 \times 1.25 = 156.25W$$

$$\text{Power in } 50\Omega \quad P_3 = 125 \times 2.5 = 312.5W$$